Ex.No.12 - Principal Component Analysis

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In [59]:
```
# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.decomposition import PCA
from sklearn.neighbors import KNeighborsClassifier, NearestNeighbors
```

In [43]:
```
# Load Dataset
df = pd.read_csv(r"D:\VI\FOML\Datasets\penguins.csv")
df = df.dropna()

print(df.shape)
df.head()
```

```
(332, 5)
```

Out[43]:

	culmen_length_mm	culmen_depth_mm	flipper_length_mm	body_mass_g	
0	39.1	18.7	181	3750	M
1	39.5	17.4	186	3800	FEM
2	40.3	18.0	195	3250	FEM
3	36.7	19.3	193	3450	FEM
4	39.3	20.6	190	3650	M

In [44]:
```
# Encode target column (sex)
le = LabelEncoder()
df["sex"] = le.fit_transform(df["sex"])

# Features and Target
X = df.drop("sex", axis=1)
y = df["sex"]

X.head()
```

Out[44]:

	culmen_length_mm	culmen_depth_mm	flipper_length_mm	body_mass_g
0	39.1	18.7	181	3750
1	39.5	17.4	186	3800
2	40.3	18.0	195	3250
3	36.7	19.3	193	3450
4	39.3	20.6	190	3650

In [45]:
```
# Standard Scaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

In [46]:
```
# Apply PCA for dimensionality reduction (4D to 2D)
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
```

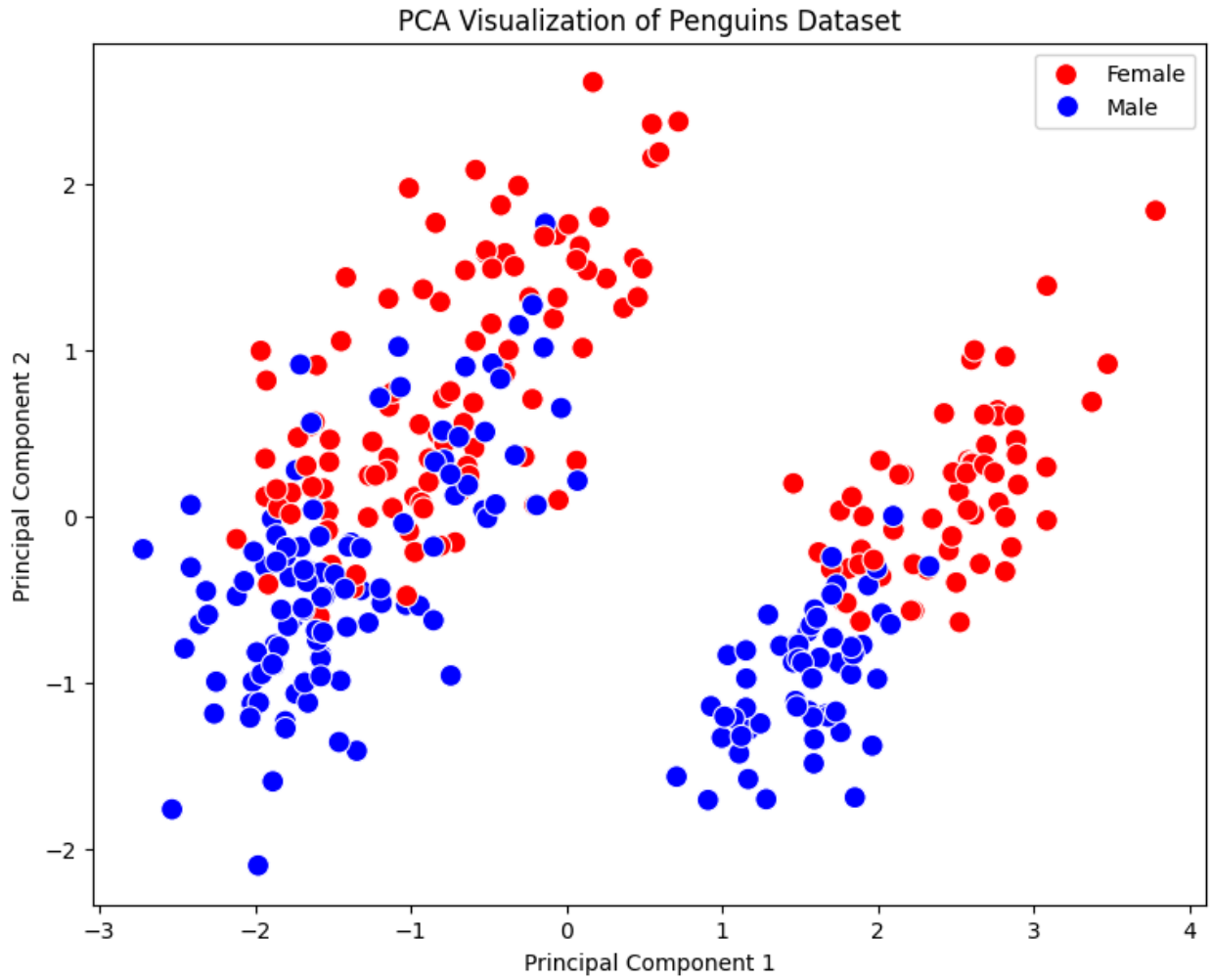
In [47]:
```
# Variance ratio
print("Explained Variance Ratio :", pca.explained_variance_ratio_)
print("Total Variance Covered :", round(sum(pca.explained_variance_ratio_) * 1
```

```
Explained Variance Ratio : [0.68713344 0.19641564]
Total Variance Covered : 88.35 %
```

In [48]:
```
# Visualize PCA result
plt.figure(figsize=(9,7))

sns.scatterplot(
    x=X_pca[:,0],
    y=X_pca[:,1],
    hue=y,
    palette={0:"blue", 1:"red"},
    s=100
)

plt.title("PCA Visualization of Penguins Dataset")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend(labels=["Female", "Male"])
plt.show()
```



In [49]:
```
# Print the loadings
loadings = pd.DataFrame(
    pca.components_.T,
    columns=["PC1", "PC2"],
    index=X.columns
)

loadings
```

Out[49]:

	PC1	PC2
culmen_length_mm	0.453174	0.604990
culmen_depth_mm	-0.398518	0.792959
flipper_length_mm	0.576880	0.003798
body_mass_g	0.550478	0.072032

In [50]:
```
# Train KNN model on PCA data
knn_pca = KNeighborsClassifier(n_neighbors=5)
knn_pca.fit(X_pca, y)
```

Out[50]:

```
▾ KNeighborsClassifier
KNeighborsClassifier()
```

In [81]:

```
# Create custom input data
custom_data = pd.DataFrame([{
    "culmen_length_mm": 45.2,
    "culmen_depth_mm": 17.1,
    "flipper_length_mm": 210,
    "body_mass_g": 4500
}])

# Scale and transform custom data using PCA
custom_scaled = scaler.transform(custom_data)
custom_pca = pca.transform(custom_scaled)

print("PC1 =", round(custom_pca[0][0], 3))
print("PC2 =", round(custom_pca[0][1], 3))
```

```
PC1 = 0.681
PC2 = 0.138
```

In [82]:

```
# Predict class of custom input
pred = knn_pca.predict(custom_pca)[0]

print("Predicted Class :", le.inverse_transform([pred])[0])
```

```
Predicted Class : MALE
```

In [83]:

```
# Plot dataset with custom point and 5 nearest neighbors
# Find 5 nearest points in PCA space
nn = NearestNeighbors(n_neighbors=5)
nn.fit(X_pca)

distances, indices = nn.kneighbors(custom_pca)

plt.figure(figsize=(9,7))

# Plot original PCA points
sns.scatterplot(
    x=X_pca[:,0],
    y=X_pca[:,1],
    hue=y,
    palette={0:"blue", 1:"red"},
    s=75
)

# Plot nearest neighbors
plt.scatter(
    X_pca[indices[0],0],
    X_pca[indices[0],1],
    c="lime",
    s=75,
```

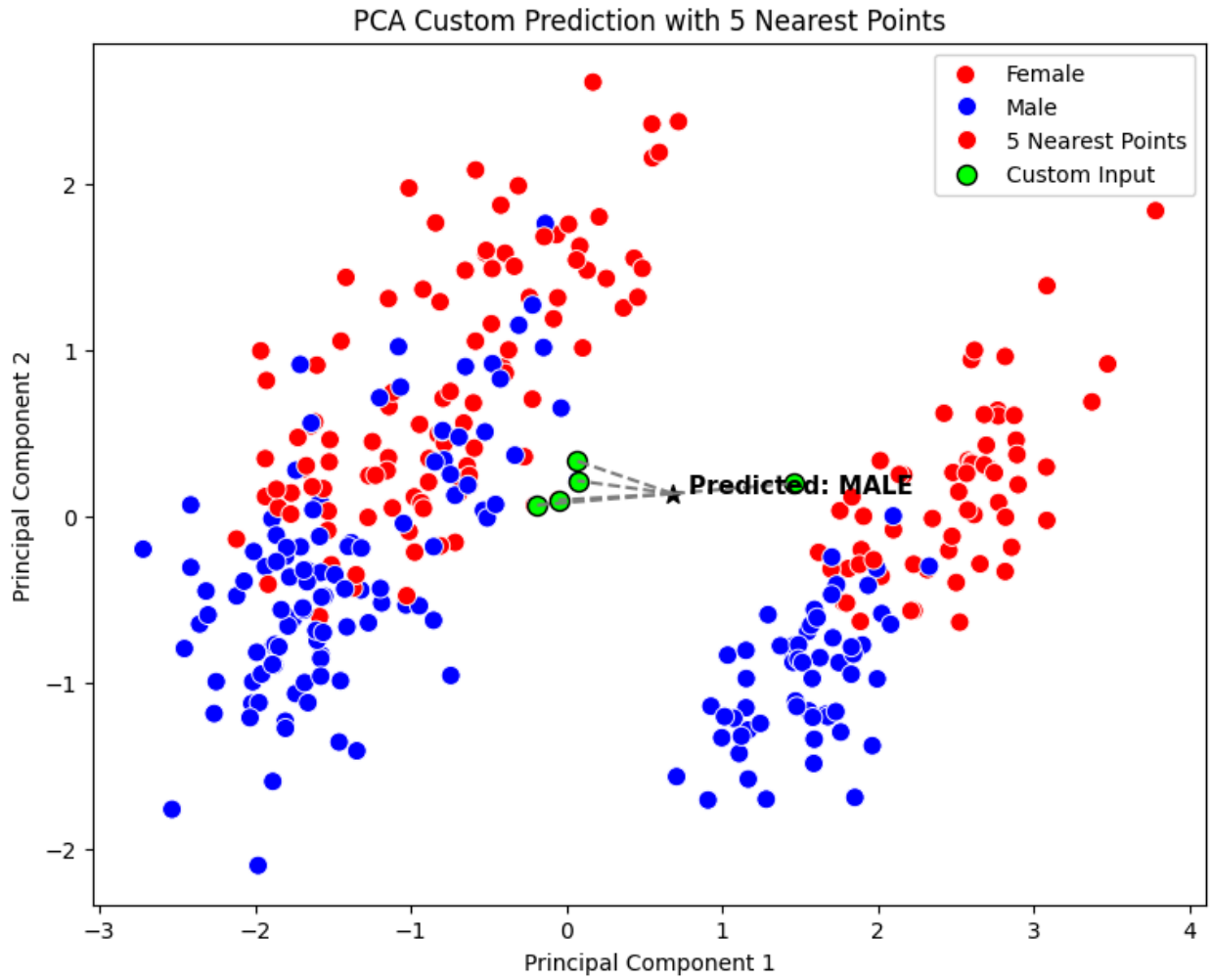
```python
    edgecolors="black",
    label="5 Nearest Points"
)

# Plot custom point
plt.scatter(
    custom_pca[:,0],
    custom_pca[:,1],
    c="black",
    marker="*",
    s=75,
    label="Custom Input"
)

# Draw lines from custom point to neighbors
for idx in indices[0]:
    plt.plot(
        [custom_pca[0,0], X_pca[idx,0]],
        [custom_pca[0,1], X_pca[idx,1]],
        '--',
        color='gray',
        linewidth=1.5
    )

# Predicted class text
plt.text(
    custom_pca[0][0] + 0.1,
    custom_pca[0][1],
    f"Predicted: {le.inverse_transform([pred])[0]}",
    fontsize=11,
    weight="bold"
)

plt.title("PCA Custom Prediction with 5 Nearest Points")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend(labels=["Female", "Male", "5 Nearest Points", "Custom Input"])
plt.show()
```


PCA Custom Prediction with 5 Nearest Points
Female
Male
5 Nearest Points
Custom Input
Predicted: MALE
Principal Component 1
Principal Component 2
−3
−2
−1
0
1
2
3
4
−2
−1
0
1
2